Voltage dividers & sensors

National 5 · Electronics & control · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Calculate V2.

- V2 = Vs × R2 ÷ (R1 + R2) = 6 × 15 ÷ 25
- V2 = 3.6 V

3 Quick check

- a) To adjust the level at which the circuit switches
- b) Falls
- c) The sensor's resistance changing
- d) 6 V

4 Key terms

- 1 – C
- 2 – A
- 3 – B
- 4 – E
- 5 – D

5 Calculate Vout across the thermistor at 20 °C.

- Vout = 5 × 2.2 ÷ (4.7 + 2.2)
- Vout = 1.6 V

6 Calculate the value of R1.

- V1 = 6 − 1.5 = 4.5 V
- R1 = R2 × V1 ÷ V2 = 1200 × 4.5 ÷ 1.5
- R1 = 3600 Ω (3.6 kΩ)

7 Calculate the resistance of R1.

- V2 = 24 − 9 = 15 V
- R1 = (V1 ÷ V2) × R2 = (9 ÷ 15) × 26
- R1 = 15.6 Ω